TORSION LINE BUNDLES DESCEND ALONG PURELY INSEPARABLE EXTENSIONS

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 1625-1635 of sources/stacks-pic/pinned-original-source.tex; proof span is lines 1637-1718, the complete author proof (80 body lines).

Editorial note. Navigation stubs: 6 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Queue index 3 of queue_pic.json (pb=3551, stmt 1625-1635, proof 1637-1718).

Editorial note. stacks-pic wave-1 (pi01): fresh adjudication accept (2026-10-14); pool per standing queue hypercovering -> pic (SMC reassigned to Zhen per Chief 08:5x); pic.tex 27 labels/23 with proof judged 21 accept / 2 reject / 4 pb=0 env-dup rows excluded; spans cite-free and reference/slogan/history-free so no patchenv needed.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

Source reference. Original lemma 0CAP: sources/stacks-pic/pinned-original-source.tex, source lines 1718–1718. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 02KH: sources/stacks-pic/pinned-original-source.tex, source lines 1718–1718. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 023B: sources/stacks-pic/pinned-original-source.tex, source lines 1718–1718. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 023T: sources/stacks-pic/pinned-original-source.tex, source lines 1718–1718. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Date: stacks-project:pic.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 0C6S: sources/stacks-pic/pinned-original-source.tex, source lines 1718–1718. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0CC5: sources/stacks-pic/pinned-original-source.tex, source lines 1718–1718. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let k be a field of characteristic $p > 0$. Let X be a quasi-compact and quasi-separated scheme over k with $H^0(X, \mathcal{O}_X) = k$. Let n be an integer prime to p. Then the map*

$$\mathrm{Pic}(X)[n] \longrightarrow \mathrm{Pic}(X_{k'})[n]$$

is bijective for any purely inseparable extension k'/k.

Proof. First we observe that the map $\mathrm{Pic}(X) \to \mathrm{Pic}(X_{k'})$ is injective by Varieties, Lemma 0CC5. Hence we have to show the map in the lemma is surjective. Let $\mathcal{L}$ be an invertible $\mathcal{O}_{X_{k'}}$-module which has order dividing n in $\mathrm{Pic}(X_{k'})$. Choose an isomorphism $\alpha : \mathcal{L}^{\otimes n} \to \mathcal{O}_{X_{k'}}$ of invertible modules. We will prove that we can descend the pair $(\mathcal{L}, \alpha)$ to X.

Set $A = k' \otimes_k k'$. Since k'/k is purely inseparable, the kernel of the multiplication map $A \to k'$ is a locally nilpotent ideal I of A. Observe that

$$X_A = X \times_{\mathrm{Spec}(k)} \mathrm{Spec}(A) = X_{k'} \times_X X_{k'}$$

comes with two projections $\mathrm{pr}_i : X_A \to X_{k'}$, $i = 0, 1$ which agree over A/I. Hence the invertible modules $\mathcal{L}_i = \mathrm{pr}_i^*\mathcal{L}$ agree over the closed subscheme $X_{A/I} = X_{k'}$. Since $X_{A/I} \to X_A$ is a thickening and since $\mathcal{L}_i$ are n-torsion, we see that there exists an isomorphism $\varphi : \mathcal{L}_0 \to \mathcal{L}_1$ by More on Morphisms, Lemma 0C6S. We may pick φ to reduce to the identity modulo I. Namely, $H^0(X, \mathcal{O}_X) = k$ implies $H^0(X_{k'}, \mathcal{O}_{X_{k'}}) = k'$ by Cohomology of Schemes, Lemma 02KH and $A \to k'$ is surjective hence we can adjust φ by multiplying by a suitable element of A. Consider the map

$$\lambda : \mathcal{O}_{X_A} \xrightarrow{\mathrm{pr}_0^*\alpha^{-1}} \mathcal{L}_0^{\otimes n} \xrightarrow{\varphi^{\otimes n}} \mathcal{L}_1^{\otimes n} \xrightarrow{\mathrm{pr}_0^*\alpha} \mathcal{O}_{X_A}$$

We can view λ as an element of A because $H^0(X_A, \mathcal{O}_{X_A}) = A$ (same reference as above). Since φ reduces to the identity modulo I we see that $\lambda = 1 \bmod I$. Then there is a unique nth root of λ in $1 + I$ (Algebra, Lemma 0CAP) and after multiplying φ by its inverse we get $\lambda = 1$. We claim that $(\mathcal{L}, \varphi)$ is a descent datum for the fpqc covering $\{X_{k'} \to X\}$ (Descent, Definition 023B). If true, then $\mathcal{L}$ is the pullback of an invertible $\mathcal{O}_X$-module $\mathcal{N}$ by Descent, Proposition 023T. Injectivity of the map on Picard groups shows that $\mathcal{N}$ is a torsion element of $\mathrm{Pic}(X)$ of the same order as $\mathcal{L}$.

Proof of the claim. To see this we have to verify that

$$\mathrm{pr}_{12}^*\varphi \circ \mathrm{pr}_{01}^*\varphi = \mathrm{pr}_{02}^*\varphi \quad \text{on} \quad X_{k'} \times_X X_{k'} \times_X X_{k'} = X_{k' \otimes_k k' \otimes_k k'}$$

As before the diagonal morphism $\Delta : X_{k'} \to X_{k' \otimes_k k' \otimes_k k'}$ is a thickening. The left and right hand sides of the equality signs are maps $a, b : p_0^*\mathcal{L} \to p_2^*\mathcal{L}$ compatible with $p_0^*\alpha$ and $p_2^*\alpha$ where $p_i : X_{k' \otimes_k k' \otimes_k k'} \to X_{k'}$ are the projection morphisms. Finally, a, b pull back to the same map under Δ. Affine locally (in local trivializations) this means that a, b are given by multiplication by invertible functions which reduce to the same function modulo a locally nilpotent ideal and which have the same nth powers. Then it follows from Algebra, Lemma 0CAP that these functions are the same. □